

Peter-Weyl theorem; principal bundle

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Problem 1

An irreducible $SU(2)$ representation of weight n has dimension $n + 1$. Moreover there is only one irreducible representation of weight n up to isomorphism. If the weight is 0, then we are considering a representation on $\mathbb{C}$. Such a representation must be the trivial representation $gz = z$, and thus the only matrix coefficient is 1. Now consider an irreducible weight 1 representation on $\mathbb{C}^2$ with basis $\{e_1, e_2\}$. Write $T(g)$ for the matrix representation of the action of g on $\mathbb{C}^2$. The representation is the canonical representation

$$gz = T(g)z = \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}.$$

Thus the matrix coefficients are precisely those matrix coefficients. Finally, consider a weight 2 representation. We consider this representation on the symmetric power $\text{Sym}^2 \mathbb{C}^2$. We use an orthonormal basis $\{e_1^2, e_2^2, \sqrt{2}e_1e_2\}$. The action of g on the basis elements is given by

$$g(e_1^2) = (ge_1)^2 = (ae_1 - \bar{b}e_2)^2 = a^2e_1^2 + \bar{b}^2e_2^2 - 2a\bar{b}e_1e_2$$

$$g(e_2^2) = (ge_2)^2 = (be_1 + \bar{a}e_2)^2 = b^2e_1^2 + \bar{a}^2e_2^2 + 2\bar{a}be_1e_2$$

$$g(\sqrt{2}e_1e_2) = \sqrt{2}(ge_1)(ge_2) = \sqrt{2}(ae_1 - \bar{b}e_2)(be_1 + \bar{a}e_2) = \sqrt{2}abe_1^2 + \sqrt{2}(a\bar{a} - b\bar{b})e_1e_2 - \sqrt{2}\bar{a}\bar{b}e_2^2.$$

Thus the matrix coefficients of g are given by the matrix representation in the $\{e_1^2, e_2^2, \sqrt{2}e_1e_2\}$ basis:

$$T(g) = \begin{pmatrix} a^2 & b^2 & \sqrt{2}ab \\ \bar{b}^2 & \bar{a}^2 & -\sqrt{2}\bar{a}\bar{b} \\ -\sqrt{2}a\bar{b} & \sqrt{2}\bar{a}b & a\bar{a} - b\bar{b} \end{pmatrix}.$$

(b) Write the matrix coefficients of g for the weight n representation as $t_{ij}^n(g)$. The Wikipedia page for Peter-Weyl says that provided our representations are unitary,

$$\left\{ \sqrt{d^\pi} u_{ij}^\pi \mid \pi \in \Sigma, 1 \leq i, j \leq d^\pi \right\}.$$

Here Σ denotes isomorphism classes of irreducible representations for a fixed weight, and d^π is the dimension of the representation. Here there is just 1 isomorphism class and all our representations are unitary, so we get the following orthonormal sets. For weight 0, we still just have $\{1\}$. For weight 1, we just scale by $\sqrt{2}$:

$$\left\{ \sqrt{2}a, \sqrt{2}b, -\sqrt{2}\bar{b}, \sqrt{2}\bar{a} \right\}.$$

For weight 2, we scale by $\sqrt{3}$:

$$\left\{ \sqrt{3}a^2, \sqrt{6}ab, \sqrt{3}b^2, -\sqrt{6}a\bar{b}, \sqrt{3}(a\bar{a} - b\bar{b}), \sqrt{6}\bar{a}b, \sqrt{3}\bar{b}^2, -\sqrt{6}\bar{a}\bar{b}, \sqrt{3}\bar{a}^2 \right\}.$$

(c) (1) The approximation of f is given by

$$f_{\leq 2}(g) = \sum_{n=0}^2 (n+1) \sum_{i=0}^n \sum_{j=0}^n \langle f, t_{ij}^n \rangle t_{ij}^n(g).$$

(2) Since $\langle f, t_{ij}^n \rangle$ is an integral, this representation suffices.

Problem 2

- (1) The polynomials given by trace $P_1 = \text{tr}$ and determinant $P_2 = \det$.
 (2) The commutator is

$$[A_\mu, A_\nu] = \begin{pmatrix} \bar{b}_\mu b_\nu - b_\mu \bar{b}_\nu & (a_\mu - d_\mu)b_\nu - (a_\nu - d_\nu)b_\mu \\ -(\bar{a}_\mu - \bar{d}_\mu)\bar{b}_\nu + (\bar{a}_\nu - \bar{d}_\nu)\bar{b}_\mu & b_\mu \bar{b}_\nu - \bar{b}_\mu b_\nu \end{pmatrix}.$$

Thus

$$F_{\mu\nu} = \begin{pmatrix} \frac{\partial_\mu a_\nu - \partial_\nu a_\mu + \bar{b}_\mu b_\nu - b_\mu \bar{b}_\nu}{\partial_\mu b_\nu - \partial_\nu b_\mu - (\bar{a}_\mu - \bar{d}_\mu)\bar{b}_\nu + (\bar{a}_\nu - \bar{d}_\nu)\bar{b}_\mu} & \frac{\partial_\mu b_\nu - \partial_\nu b_\mu + (a_\mu - d_\mu)b_\nu - (a_\nu - d_\nu)b_\mu}{\partial_\mu d_\nu - \partial_\nu d_\mu + b_\mu \bar{b}_\nu - \bar{b}_\mu b_\nu} \end{pmatrix}.$$

Write these matrix components as

$$F_{\mu\nu} = \begin{pmatrix} F_{\mu\nu}^{11} & F_{\mu\nu}^{12} \\ F_{\mu\nu}^{21} & F_{\mu\nu}^{22} \end{pmatrix}.$$

Then we have

$$F_A = \frac{1}{2} \begin{pmatrix} F_{\mu\nu}^{11} & F_{\mu\nu}^{12} \\ F_{\mu\nu}^{21} & F_{\mu\nu}^{22} \end{pmatrix} dx^\mu \wedge dx^\nu.$$

Then $\text{tr}(F_A)$ is

$$\text{tr}(F_A) = (F_{\mu\nu}^{11} + F_{\mu\nu}^{22}) dx^\mu \wedge dx^\nu = \frac{1}{2} (\partial_\mu a_\nu - \partial_\nu a_\mu + \partial_\mu d_\nu - \partial_\nu d_\mu) dx^\mu \wedge dx^\nu.$$

A similar computation shows

$$\text{tr}(F_A^2) = \frac{1}{4} (F_{\mu\nu}^{11} F_{\rho\sigma}^{11} + F_{\mu\nu}^{22} F_{\rho\sigma}^{22} + 2F_{\mu\nu}^{12} F_{\rho\sigma}^{21}) dx^\mu \wedge dx^\nu \wedge dx^\rho \wedge dx^\sigma.$$

This provides explicit formulas

$$c_1(A) = \frac{i}{2\pi} (F_{\mu\nu}^{11} + F_{\mu\nu}^{22}) dx^\mu \wedge dx^\nu,$$

$$c_2(A) = \frac{1}{16\pi^2} (F_{\mu\nu}^{11} F_{\rho\sigma}^{22} - F_{\mu\nu}^{12} F_{\rho\sigma}^{21}) dx^\mu \wedge dx^\nu \wedge dx^\rho \wedge dx^\sigma.$$

(3) The Bianchi identity states that the covariant derivative of the curvature $d_A F_A = 0$. In terms of the exterior derivative, this states

$$dF_A + [A, F_A] = 0 \implies dF_A = F_A \wedge A - A \wedge F_A.$$

We also use the graded commutativity of the trace for Lie algebra-valued forms. For $\alpha \in \Omega^k(M, \mathfrak{u}(2))$ and $\beta \in \Omega^l(M, \mathfrak{u}(2))$, we have $\text{tr}(\alpha \wedge \beta) = (-1)^{kl} \text{tr}(\beta \wedge \alpha)$. Specifically, if one of the forms has even degree, the trace of their wedge products commutes.

For $c_1(A)$, taking the exterior derivative yields:

$$d(\text{tr}(F_A)) = \text{tr}(dF_A) = \text{tr}(F_A \wedge A - A \wedge F_A) = \text{tr}(F_A \wedge A) - \text{tr}(A \wedge F_A).$$

Since F_A is a 2-form, $\text{tr}(F_A \wedge A) = \text{tr}(A \wedge F_A)$, and thus $d(\text{tr}(F_A)) = 0$. Consequently, $dc_1(A) = 0$.

For $c_2(A)$, we must show that $\text{tr}(F_A^2)$ is closed, since we already know $(\text{tr}(F_A))^2$ is closed. Applying the exterior derivative and the Leibniz rule:

$$d(\text{tr}(F_A \wedge F_A)) = \text{tr}(dF_A \wedge F_A + F_A \wedge dF_A) = 2\text{tr}(dF_A \wedge F_A).$$

Substituting the Bianchi identity gives:

$$d(\text{tr}(F_A^2)) = 2\text{tr}([F_A, A] \wedge F_A) = 2\text{tr}(F_A \wedge A \wedge F_A - A \wedge F_A \wedge F_A).$$

Since $F_A \wedge A$ is a 3-form and F_A is a 2-form, their trace commutes, meaning $\text{tr}(F_A \wedge A \wedge F_A) = \text{tr}(A \wedge F_A \wedge F_A)$. Thus, the right-hand side vanishes, giving $d(\text{tr}(F_A^2)) = 0$. It follows that $dc_2(A) = 0$.

Let A_t be a smooth family of connections on P . The derivative $\dot{A}_t = \frac{d}{dt}A_t$ is a globally defined $\mathfrak{u}(2)$ -valued 1-form. The curvature of A_t is $F_{A_t} = dA_t + A_t \wedge A_t$. Differentiating this with respect to t yields:

$$\frac{d}{dt}F_{A_t} = d\dot{A}_t + \dot{A}_t \wedge A_t + A_t \wedge \dot{A}_t = d\dot{A}_t + [A_t, \dot{A}_t] = d_{A_t}\dot{A}_t.$$

For $i = 1$, we evaluate the derivative of $\text{tr}(F_{A_t})$:

$$\frac{d}{dt}\text{tr}(F_{A_t}) = \text{tr}(d_{A_t}\dot{A}_t) = \text{tr}(d\dot{A}_t + [A_t, \dot{A}_t]) = d(\text{tr}(\dot{A}_t)) + \text{tr}([A_t, \dot{A}_t]).$$

Since A_t and $\dot{A}_t$ are both 1-forms, their commutator trace vanishes: $\text{tr}(A_t \wedge \dot{A}_t + \dot{A}_t \wedge A_t) = \text{tr}(A_t \wedge \dot{A}_t) - \text{tr}(A_t \wedge \dot{A}_t) = 0$. Thus,

$$\frac{d}{dt}c_1(A_t) = d\left(\frac{i}{2\pi}\text{tr}(\dot{A}_t)\right),$$

which is globally exact.

For $i = 2$, we differentiate the components of $c_2(A_t)$. First, for the squared trace term:

$$\frac{d}{dt}(\text{tr}(F_{A_t}))^2 = 2\text{tr}(F_{A_t}) \wedge \frac{d}{dt}\text{tr}(F_{A_t}) = 2\text{tr}(F_{A_t}) \wedge d(\text{tr}(\dot{A}_t)).$$

Because $\text{tr}(F_{A_t})$ is closed, we can pull the exterior derivative to the outside:

$$\frac{d}{dt}(\text{tr}(F_{A_t}))^2 = d\left(2\text{tr}(\dot{A}_t) \wedge \text{tr}(F_{A_t})\right),$$

which is exact. Next, we differentiate $\text{tr}(F_{A_t}^2)$:

$$\frac{d}{dt}\text{tr}(F_{A_t} \wedge F_{A_t}) = 2\text{tr}\left(\frac{d}{dt}F_{A_t} \wedge F_{A_t}\right) = 2\text{tr}(d_{A_t}\dot{A}_t \wedge F_{A_t}).$$

Expanding the covariant derivative:

$$\text{tr}(d_{A_t}\dot{A}_t \wedge F_{A_t}) = \text{tr}(d\dot{A}_t \wedge F_{A_t}) + \text{tr}([A_t, \dot{A}_t] \wedge F_{A_t}).$$

Consider the exterior derivative of the 3-form $\text{tr}(\dot{A}_t \wedge F_{A_t})$:

$$d(\text{tr}(\dot{A}_t \wedge F_{A_t})) = \text{tr}(d\dot{A}_t \wedge F_{A_t}) - \text{tr}(\dot{A}_t \wedge dF_{A_t}) = \text{tr}(d\dot{A}_t \wedge F_{A_t}) - \text{tr}(\dot{A}_t \wedge [F_{A_t}, A_t]).$$

Expanding the commutator traces yields $\text{tr}([A_t, \dot{A}_t] \wedge F_{A_t}) = \text{tr}(A_t \wedge \dot{A}_t \wedge F_{A_t} + \dot{A}_t \wedge A_t \wedge F_{A_t})$. This is equal to $-\text{tr}(\dot{A}_t \wedge (F_{A_t} \wedge A_t - A_t \wedge F_{A_t})) = -\text{tr}(\dot{A}_t \wedge [F_{A_t}, A_t])$. Thus,

$$\text{tr}(d_{A_t}\dot{A}_t \wedge F_{A_t}) = d(\text{tr}(\dot{A}_t \wedge F_{A_t})),$$

meaning $\frac{d}{dt}\text{tr}(F_{A_t}^2) = 2d(\text{tr}(\dot{A}_t \wedge F_{A_t}))$, which is exact. Since both components of the derivative of $c_2(A_t)$ are exact, $\frac{d}{dt}c_2(A_t)$ is exact.

Since $\frac{d}{dt}c_i(A_t) = d\alpha_{i,t}$ for some globally defined forms $\alpha_{i,t}$, we integrate with respect to t from 0 to 1:

$$c_i(A_1) - c_i(A_0) = \int_0^1 \frac{d}{dt}c_i(A_t)dt = \int_0^1 d\alpha_{i,t}dt = d\left(\int_0^1 \alpha_{i,t}dt\right).$$

Because the difference $c_i(A_1) - c_i(A_0)$ is an exact form, it maps to zero in de Rham cohomology. Therefore, the cohomology classes $[c_i(A_0)] = [c_i(A_1)] \in H_{dR}^{2i}(M)$ are identical.

(4) The trace is $\omega_1 + \omega_2$. Thus

$$c_1(A) = \frac{i}{2\pi}(\omega_1 + \omega_2).$$

Then

$$F_A = \begin{pmatrix} \omega_1 \wedge \omega_1 & 0 \\ 0 & \omega_2 \wedge \omega_2 \end{pmatrix}.$$

We also have

$$(\text{tr}(F_A))^2 = \omega_1 \wedge \omega_1 + \omega_2 \wedge \omega_2 + 2\omega_1 \wedge \omega_2.$$

Thus $\text{tr}(F_A^2) - (\text{tr}(F_A))^2 = -2\omega_1 \wedge \omega_2$. Thus

$$c_2(A) = -\frac{1}{8\pi^2}(-2\omega_1 \wedge \omega_2) = \frac{1}{4\pi}(\omega_1 \wedge \omega_2).$$